JACOB BARRETT

Technical Sound Designer & Game Developer
Christchurch, New Zealand | Bachelor of Digital Screen (Honours) — Major in Game Development (University of Canterbury, 2026)
Portfolio

PROFESSIONAL SUMMARY

Specialized Technical Sound Designer and Game Developer with expertise bridging interactive audio systems and gameplay mechanics. Skilled in implementing dynamic audio middleware (Wwise), designing real-time spatialization networks, acoustic attenuation, and gameplay scripting in C# and C++. Experienced in balancing high-stakes technical delivery across co-op game pipelines and live theatrical productions.

MAJOR PROJECT SPOTLIGHT: HEART AND METAL

Role: Technical Sound Designer & Audio Lead
Genre / Platform: Interactive 3D Puzzle / Action Title (Unity Engine)
Project Overview: A collaborative game project featuring rich environmental interactions, mechanical puzzles, and character-driven gameplay.

Key Audio Implementation & Technical Achievements

- **Spatial Audio Architecture & Wwise Rooms/Portals:** Designed and integrated Wwise Spatial Audio using Rooms and Portals to calculate dynamic room tone reverberation and environmental acoustic occlusion across complex level geometry.
- **Raycast-Driven Surface & Footstep Detection:** Implemented custom C# surface detection scripts using raycasting to dynamically trigger distinct footstep and movement audio layers based on environmental textures and terrain.
- **Split-Listener & Multi-Character Audio Routing:** Configured dual/split listener structures and sub-busses to dynamically route individual character audio streams and environmental emitters cleanly without audio masking.
- **Atmospheric & Interactive Soundscapes:** Authored custom interactive soundscapes and spot effects—including environmental ambiences (water, cinderblock movement, dirt rustle, lift mechanisms) and interactive puzzle audio (laser receivers, fans, weak structure collapses).

CORE TECHNICAL COMPETENCIES

Category	Competencies & Toolsets
Audio Middleware & Spatialization	Wwise, Spatial Audio Rooms & Portals, Acoustic Attenuation Curves, Spatialized Reverb Routing, Sub-Bussing
Languages & Scripting	C# (Advanced), C++, Python, JavaScript, HTML, CSS
Game Engines & Tools	Unity (URP, VR Interaction Toolkits), Git Version Control, JetBrains Rider
Live Production & Field Work	Live Sound Mixing, Technical Theatre Direction, Signal Routing, Field Recording

OTHER KEY PROJECTS & EXPERIENCE

- **Streetside (Co-op Escape Game):** Implemented multi-pass screen-space shaders, raycast stencils, and split-ear dual-listener spatialization networks.
- **VR Interaction Lab (Unity):** Developed interactive puzzle mechanics utilizing native VR interaction toolkits and event-driven trigger architectures.
- **Live Technical Production (MUSOC / UCSA):** Technical Director and Sound Engineer for live musical theatre productions, managing live mixing and signal flow.